Convergence Analysis and Recent Improvements of the AAA Algorithm

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1. Introduction: The Background and the Gap of the AAA Algorithm

Background

- Introduced by Nakatsukasa et al. in 2018, the AAA (adaptive Antoulas–Anderson) algorithm is a breakthrough in rational approximation.
- **Key Advantages:** Remarkable speed, robustness, and flexibility.
- **Applications:** Analytic continuation, signal processing, model order reduction, and nonlinear eigenvalue problems.

The Theoretical Gap

- Despite practical success, theoretical understanding of its convergence behavior lags significantly behind.
- Unlike traditional optimization-based methods, AAA does not seek to satisfy strict optimality conditions.

1. Introduction: Theoretical Challenges

Why is the AAA algorithm hard to analyze?

1. **Selection of Convergence Definition:** The algorithm will definitely stop on a finite number of sample points, so how to give a suitable definition of convergence has become a problem.
2. **Problem-Dependent Convergence:**
   - For analytic functions, it exhibits *super-exponential convergence*.
   - For functions with branch points, cusps, or non-differentiability, convergence slows dramatically or stalls.
3. **Denominator Oscillation:** When the denominator varies significantly, the discrepancy between the linearized error and the true nonlinear error leads to unstable convergence.

1. Introduction: Defining Convergence

The Dilemma of Convergence Definition

- On a discrete set of points, the algorithm is guaranteed to terminate in finite steps. Thus, discussing "convergence" on discrete points is trivial.
- The real challenge: Does the sequence of rational approximations $\{r_m\}$ approximate the target function f in a strong sense on a continuous domain E?

Obstacles on Continuous Domains:

- **Froissart Doublets:** Spurious pole-zero pairs that nearly cancel out but cause local spikes to infinity.
- **Unexpected Poles:** AAA does not mathematically guarantee that no poles will appear within the approximation domain.

2. Core of AAA: Problem Setup and Notation

Problem Setup

- Let $Z = \{z_1, \ldots, z_M\} \subset E \subset \mathbb{C}$ be a finite set of **sample points** ($M \geq 1$).
- **Target function** f is given at least on Z, either via an analytic expression or as discrete values $f(z_i)$.
- The algorithm constructs a sequence of **rational approximants** r_m of increasing degree.

Notation

- **Support points** $\{z_1, \ldots, z_m\} \subset Z$.
- Let the **non-support points** at iteration m be denoted as:

$$Z^{(m)} := Z \setminus \{z_1, \ldots, z_m\}$$

2. Core of AAA: Barycentric Rational Representation

The approximation function r is represented in **barycentric form** as a quotient of two partial-fraction sums:

Barycentric Formula

$$r(z) = \frac{n(z)}{d(z)} = \frac{\sum_{j=1}^{m} \frac{w_j f_j}{z - z_j}}{\sum_{j=1}^{m} \frac{w_j}{z - z_j}}$$

- $z_1, \ldots, z_m$ are distinct support points, $f_j = f(z_j)$, and w_j are weights.
- **Interpolation Property:** Although it exhibits ∞/∞ at $z = z_j$, the singularity is removable, ensuring $r(z_j) = f_j$ as long as $w_j \neq 0$.
- Defines a rational function of type $(m-1, m-1)$.

2. Core of AAA: Greedy Point Selection & Linearization

Greedy Support Point Selection:

- At step m, the new support point z_m is chosen from $Z^{(m-1)}$ where the nonlinear residual $|f(z) - r_{m-1}(z)|$ reaches its maximum.

Linearization:

- The goal is $f(z) \approx \frac{n(z)}{d(z)}$ for $z \in Z$.
- AAA linearizes this strictly on the non-support set $Z^{(m)}$:

$$f(z)d(z) \approx n(z), \quad z \in Z^{(m)}$$

- Cannot evaluate on support points directly because $n(z)$ and $d(z)$ have poles at z_j.

2. Core of AAA: The Least-Squares Problem

The weights $w = (w_1, \ldots, w_m)^T$ are chosen to minimize the discrete 2-norm of the linearized residual with normalization $||w||_2 = 1$:

Minimization Problem

$$\min_{||w||_2=1} ||fd - n||_{\ell^2(Z^{(m)})}$$

Matrix Formulation: Let $F^{(m)} = f(Z^{(m)})$. This reduces to:

$$\min_{||w||_2=1} ||A^{(m)}w||_2$$

where $A^{(m)} \in \mathbb{C}^{(M-m)\times m}$ is the Loewner matrix:

$$A^{(m)}_{i,j} = \frac{F^{(m)}_i - f_j}{Z^{(m)}_i - z_j}$$

2. Core of AAA: Solving via SVD

To solve the matrix problem $\min_{||w||_2=1} ||A^{(m)}w||_2$:

- Take the thin SVD: $A^{(m)} = U\Sigma V^*$, where $\Sigma = \text{diag}(\sigma_1, \dots, \sigma_m)$ with $\sigma_1 \geq \cdots \geq \sigma_m \geq 0$.
- Let $w = Vy$. Since V is unitary, $||w||_2 = ||y||_2 = 1$.
- Since U has orthonormal columns, $||U\Sigma y||_2 = ||\Sigma y||_2$.

Reduced Problem

$$\min_{||y||_2=1} ||\Sigma y||_2 = \min_{||y||_2=1} \sum_{j=1}^{m} \sigma_j^2 |y_j|^2$$

- The minimum is achieved at $y = e_m$.
- The minimizer is the last right singular vector: $w = v_m$.

2. Core of AAA: Efficient Matrix Updates

For efficient evaluation, AAA avoids building the full Loewner matrix from scratch.

- Uses the Cauchy matrix $C^{(m)} \in \mathbb{C}^{(M-m)\times m}$: $C_{i,j}^{(m)} = \frac{1}{Z_i^{(m)} - z_j}$.
- Loewner matrix structure: $A^{(m)} = S_F C - C S_f$ (where $S_F = diag(F_1^{(m)}, \ldots, F_{M-m}^{(m)})$ and $S_f = diag(f_1, \ldots, f_m)$ are diagonal matrices of function values).

Once w is found, vectors for numerator N and denominator D are:

$$N = C(w \odot \vec{f}), \quad D = Cw$$

Where $\odot$ is the Hadamard product. Finally, approximation values on $Z^{(m)}$ are updated as $R(Z^{(m)}) = N./D$.

2. Core of AAA: Stopping Criterion and Cleanup

Stopping Criterion: Iteration stops when relative nonlinear residual falls below a tolerance (typically 10^{-13}):

$$\max_{z\in Z^{(m)}} |f(z) - r_m(z)| \leq \text{tol} \cdot \max_{z\in Z} |f(z)|$$

Cleanup Strategy for Froissart Doublets:

- Approximations often contain pole-zero pairs that nearly cancel out, accompanied by extremely small residues.
- These poles cause the function to approach infinity locally, destroying uniform convergence.
- **Mechanism:** Inspect computed poles. If a residue's magnitude is below a threshold (e.g., 10^{-13}), it is flagged as spurious.
- **Cleanup:** Remove the support point z_j nearest to the spurious pole, and re-solve the SVD over the reduced support set.

3. Convergence: Three Notions of Convergence

We explore three frameworks to analyze the convergence of AAA:

1. **Termination on a Finite Set**: Guaranteed, but weak.
2. **Uniform Convergence**: Desirable, but often too strict.
3. **Convergence in Measure**: Weaker, but highly suitable for handling local poles.

3. Convergence: The First Two Types of Convergence

1. Termination of the AAA Iteration

- Since support points are chosen from a finite set Z, the available points are exhausted eventually.
- However, a small error on Z does not guarantee a good approximation on the continuous domain E.

2. Uniform Convergence ($r_m \to f$ uniformly on E)

- **Definition**: $\forall \epsilon > 0, \exists m_0$ such that $||f - r_m||_{L^\infty(E)} < \epsilon$ for all $m \geq m_0$.
- **Why it fails for AAA:** Too strong to serve as a baseline. $||f - r_m||_{\ell^\infty(Z)}$ can be small while $||f - r_m||_{L^\infty(E)}$ is infinite due to isolated poles (Froissart doublets or inherent singularities) falling *between* sample points.

3. Convergence: Convergence in Measure

Because isolated poles disrupt uniform convergence locally without ruining the overall approximation, we turn to **Convergence in Measure**.

Definition: Convergence in Measure

For continuous domain E with measure μ and measurable function f:

$$\forall \alpha > 0, \forall \epsilon > 0, \exists N, \forall m > N, \quad \mu(z \in E : |f(z) - r_m(z)| > \alpha) < \epsilon$$

.

Two Main Difficulties to Prove This:

1. Handling the poles (carving them out).
2. Bridging the gap between the discrete sampled error and the continuous domain error (requires assumptions on point distribution).

3. Convergence in Measure: Dealing with Poles

Step 1: Carving out the bad set.

- The number of poles is finite. We can carve out small open neighborhoods $U_{m,\epsilon}$ around the set of poles P_m such that:

$$\forall \epsilon > 0, \quad \mu(U_{m,\epsilon}) < \frac{\epsilon}{2}$$

- We define the "good set" as:

$$G_{m,\epsilon} = E \setminus U_{m,\epsilon}$$

- On $G_{m,\epsilon}$, the denominator of r_m is strictly non-zero, making the approximant r_m completely smooth.

3. Convergence in Measure: Lipschitz & Duality

Step 2: Bounding continuous error using discrete error.

- Assume f is Lipschitz. Introduce a Lipschitz cutoff function $0 \leq \chi_m \leq 1$ that vanishes near the poles and equals 1 on the good region.
- Define $g_m(z) = |f(z) - r_m(z)|\chi_m(z)$. Let $L_m = \text{Lip}(g_m)$.
- Take μ as a Borel probability measure and let discrete empirical measure be $\mu_M = \frac{1}{M}\sum \delta_{z_i}$. Assume $L_m > 0$, define $\varphi = g_m/L_m$, then $\text{Lip}(\varphi) \leq 1$, Using Kantorovich-Rubinstein duality for Wasserstein-1 distance W_1:

$$W_1(\mu, \mu_M) = \sup_{\text{Lip}(\varphi)\leq 1} \left| \int_E \varphi \, d\mu - \int_E \varphi \, d\mu_M \right| \geq \left| \int_E \varphi \, d\mu - \int_E \varphi \, d\mu_M \right|$$

Multiply by L_m yields:

$$\left| \int_E g_m d\mu - \int_E g_m d\mu_M \right| \leq L_m W_1(\mu, \mu_M)$$

3. Convergence in Measure: Lipschitz & Duality

If $L_m = 0$, then g_m can only be 0 and the estimate is trivial.
Therefore, we can always obtain:

Measure Distance Bound

$$\left| \int_E g_m d\mu - \int_E g_m d\mu_M \right| \leq L_m W_1(\mu, \mu_M).$$

This implies the continuous integral is bounded by the discrete error plus a grid-density term:

$$\begin{aligned}
\int_E g_m \, d\mu &\leq \int_E g_m \, d\mu_M + L_m \, W_1(\mu, \mu_M) \\
&\leq \|g_m\|_{\ell^\infty(Z)} + L_m \, W_1(\mu, \mu_M) \\
&= ||g_m||_{\ell^\infty(Z^{(m)})} + L_m \, W_1(\mu, \mu_M) \\
&\leq ||f - r_m||_{\ell^\infty(Z^{(m)})} + L_m \, W_1(\mu, \mu_M).
\end{aligned}$$

Since there is $0 \leq \chi_m \leq 1$, the last inequality is true.

3. Convergence in Measure: Finalizing the Proof

Step 3: Applying Markov's Inequality.

At the same time, based on $\chi_m = 1$ on $G_{m,\epsilon}$, then

$$\forall \alpha > 0, \{|f - r_m| > \alpha\} \cap G_{m,\epsilon} \subset \{g_m > \alpha\}.$$

Using the Markov's inequality

$$\begin{aligned} \mu(\{|f - r_m| > \alpha\} \cap G_{m,\epsilon}) &\leq \mu(\{g_m > \alpha\}) \\ &\leq \frac{1}{\alpha} \int_E g_m d\mu \\ &\leq \frac{1}{\alpha}(||f - r_m||_{\ell^\infty(Z^{(m)})} + L_m W_1(\mu, \mu_M)). \end{aligned}$$

- If the sampling grid is dense enough such that $W_1(\mu, \mu_M) \leq \frac{\alpha}{4L_m}\epsilon$.
- And the algorithm stopping tolerance is set to $\eta_{m,M} < \frac{\alpha}{4}\epsilon$.
- Then $\mu(\{|f - r_m| > \alpha\} \cap G_{m,\epsilon}) < \frac{\epsilon}{2}$.

Conclusion: $\mu(z \in E : |f(z) - r_m(z)| > \alpha) \leq \mu(U_{m,\epsilon}) + \mu(\{|f - r_m| > \alpha\} \cap G_{m,\epsilon}) < \epsilon$

The Caveat: The Lipschitz constant L_m may increase rapidly with m due to denominator oscillations. Therefore, W_1 may not satisfy the equality if the sample point is selected from the beginning.

4. Limitations of the Traditional AAA

Motivated by the convergence analysis, three fundamental limitations emerge when applying traditional AAA to continuum problems:

1. **Fixed Sample Set :** Chosen a priori, its density bottlenecks the achievable accuracy and directly limits our ability to control $L_m W_1$.
2. **Distorted Objective Function:** The SVD minimizes the *linearized* residual $||fd - n||$. When f changes sharply, the denominator $d(z)$ oscillates wildly to compensate, causing L_m to explode.
3. **Near-Best, Not Minimax:** AAA is an interpolation method and there may be Runge phenomenon. In approximation theory, the effect of this approximation is not as good as that of minimax, i.e., best uniform approximation.

4. Improvements: The Continuum AAA Algorithm

Solution to Fixed Grid Limitation (Driscoll et al., 2023)

- **Dynamic Grid Refinement:** Eliminates the need for user-provided discrete sets. The user only specifies the continuous domain boundaries.
- As iteration progresses, sample points are dynamically added and locally refined around the newly selected support points.
- **Bad Pole Detection:** Evaluates pole locations at every step. If poles fall inside E, the solution is rejected. If this happens for 10 consecutive steps, it terminates and returns the last clean approximation.

Impact: Prevents poles in the domain and keeps W_1 arbitrarily small locally, guaranteeing measure-based convergence.

4. Improvements: The NL-AAA Algorithm

Solution to Distorted Objective (Ackermann et al., 2026)

- **True Nonlinear Optimization:** Traditional AAA may select the wrong support point when small denominators artificially amplify numerator errors. NL-AAA fixes this weight-solving step.
- **Inner Iterations:** Applies *Sanathanan-Koerner iterations* and *Whitfield's iterations* to refine the weights towards the true nonlinear optimal solution at each step.
- **Conservative Fallbacks:** If true error does not decrease, it zeroes out the last bit of the weight and switches greedy selection to random or relative-error-based selection.

Impact: Avoids excessive oscillation of the denominator, tightly controlling the growth of L_m.

4. Improvements: The AAA-Lawson Algorithm

Solution for Strict Minimax Optimality (Nakatsukasa & Trefethen, 2019)

- **Goal:** Minimize the maximum error across the domain, upgrading from "near-best" to "minimax".
- **Two-Stage Framework:**
  1. **Initialization:** Use standard AAA to quickly find a near-best initial guess.
  2. **Lawson Iteration:** Switches to a non-interpolated barycentric form. This completely decouples the numerator and denominator coefficients. The iteration updates sample point weights based on the current approximation error, and uses the weights obtained in the first stage as a basis for initialization.

Impact: Help reach minimax approximation.

5. Future Work and Reference

Future Work:

- Combine the given algorithmic improvements.
- Provide strict mathematical proofs bounding the convergence speed for these hybrid algorithms.

Reference

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Thank You!

Questions & Discussion